

NAME: _____ (pledged)

POINTS: _____ / 10

Pepquiz: STL vector class (in class, self-graded)

1. Your source file `main.cpp` is located in the folder `/home/src/`. Inside that same project you have created your own header file named `Rectangle.h`, which is located in `/home/include/`. Write two different valid `#include` statements that will correctly include `Rectangle.h` from inside `main.cpp` - first using an **absolute**, second using a **relative** path to the header file.
2. Write a statement that defines an empty `vector` object named `avect` that can hold strings. Show that the vector is actually empty.
3. Write a statement that defines a `vector` object named `avect` that can hold integers. The `vector` should have ten elements initialized with the default value 0. Display the elements to show success.
4. Write a statement that defines a `vector` object named `avect` that can hold integers. The `vector` should have 100 elements, each initialized with the value 1. If this worked, print a success message, but if any value is not 1, abort.
5. Write a statement that defines a `vector` object named `v1` that can hold integers. The vector should be a copy of another vector `v2`. Display both vectors to show that the copying worked.
6. What happens when you use an invalid index with the `at` function of the `vector` class?

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7. If your program will add a lot of objects to a `vector`, is it best to use the `insert` member function, or the `emplace` member function, and why?
8. Internally, how does a `vector` store its elements?
9. The `vector` class has a `size` and a `capacity` member function. What is the difference between these two member functions?
10. What is the difference between the `vector` class member functions `insert` and `push_back`? Show it with a short program.